Career Overview: Prof. Dr. Saraju P. Mohanty

A distinguished academic leader, researcher, and innovator with over two decades of experience spanning AI institute leadership, research excellence, and global IEEE governance. From founding the Smart Electronic Systems Laboratory at UNT to serving as Founding Dean of India's first AI-Native University, this career represents a remarkable journey of impact.

SRM UNIVERSITY – AP

UNIVERSITY OF NORTH TEXAS

IEEE LEADERSHIP

Career Timeline at a Glance

1
1995–2003
B.Tech (OUAT), M.E. (IISc Bangalore), Ph.D. (Univ. of South Florida)

2
2004–2010
Assistant Professor & Founding SESL Director, UNT

3
2010–2015
Associate Professor (Tenured), UNT

4
2015–Present
Full Professor (Tenured), Personnel Affairs Chair, Associate Chair/Head (in 2016) & SESL Director, UNT

5
Apr 2026–Present
Founding Dean, AI Technology Institute (AiTI), SRM University – AP

From B.Tech (OUAT) and M.E. (IISc Bangalore) through Ph.D. (USF) to Full Professor at UNT and Founding Dean of AiTI — a continuous arc of academic leadership and innovation spanning three decades.

Educational Background

Ph.D. – Computer Science & Engineering
University of South Florida, Tampa — Dec 2003

M.E. – Artificial Intelligence
Indian Institute of Science, Bangalore — Mar 1999
(formerly System Science & Automation)

B.Tech – Electrical Engineering
Orissa University of Agriculture & Technology,
Bhubaneswar — May 1995

Founding Dean: AI Technology Institute (AiTI)

SRM UNIVERSITY – AP, INDIA

APR 2026 – PRESENT

As Founding Dean, the mission is to build AiTI into India's **first AI-Native University** — one that offers core AI programs, certifies every graduating student's AI working competency, integrates AI across all disciplines, provides in-house AI computing infrastructure, and incorporates AI in all aspects of academia.

AI Programs

Core B.Tech, M.Tech & Ph.D. offerings

AI Competency

Certified for every graduate

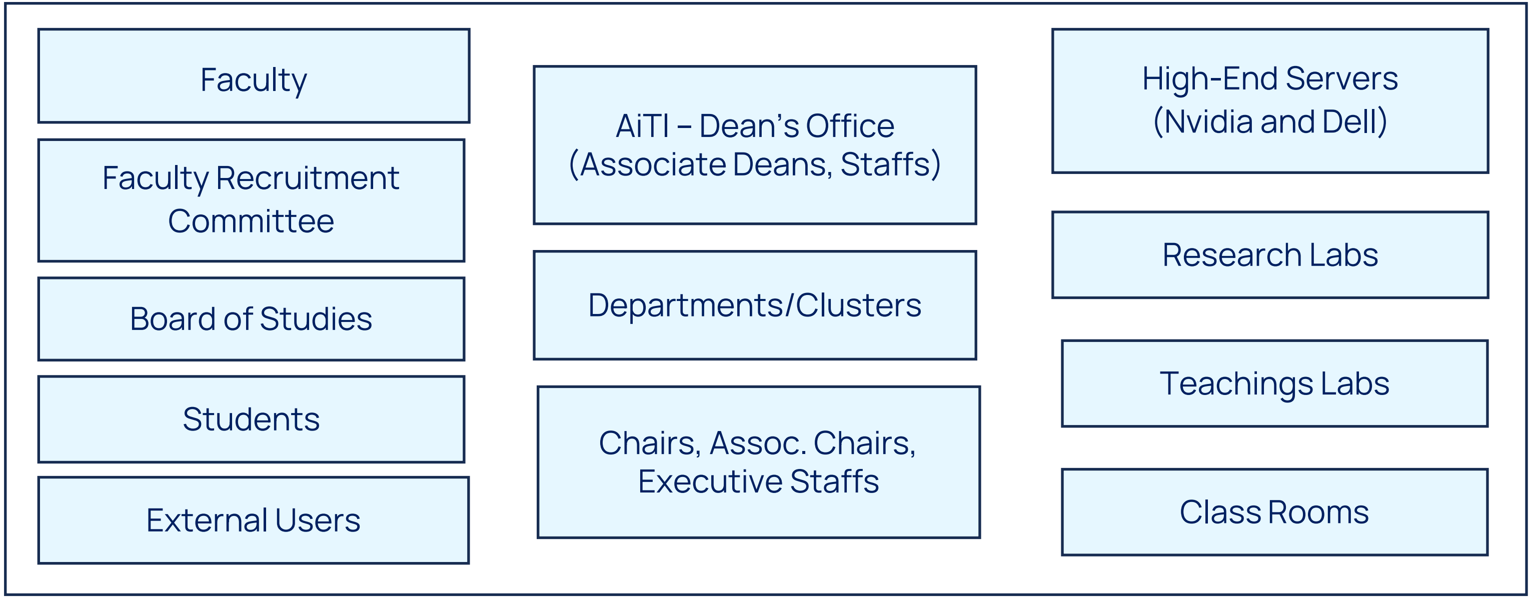
AI Integration

Across all university programs

HPC Infrastructure

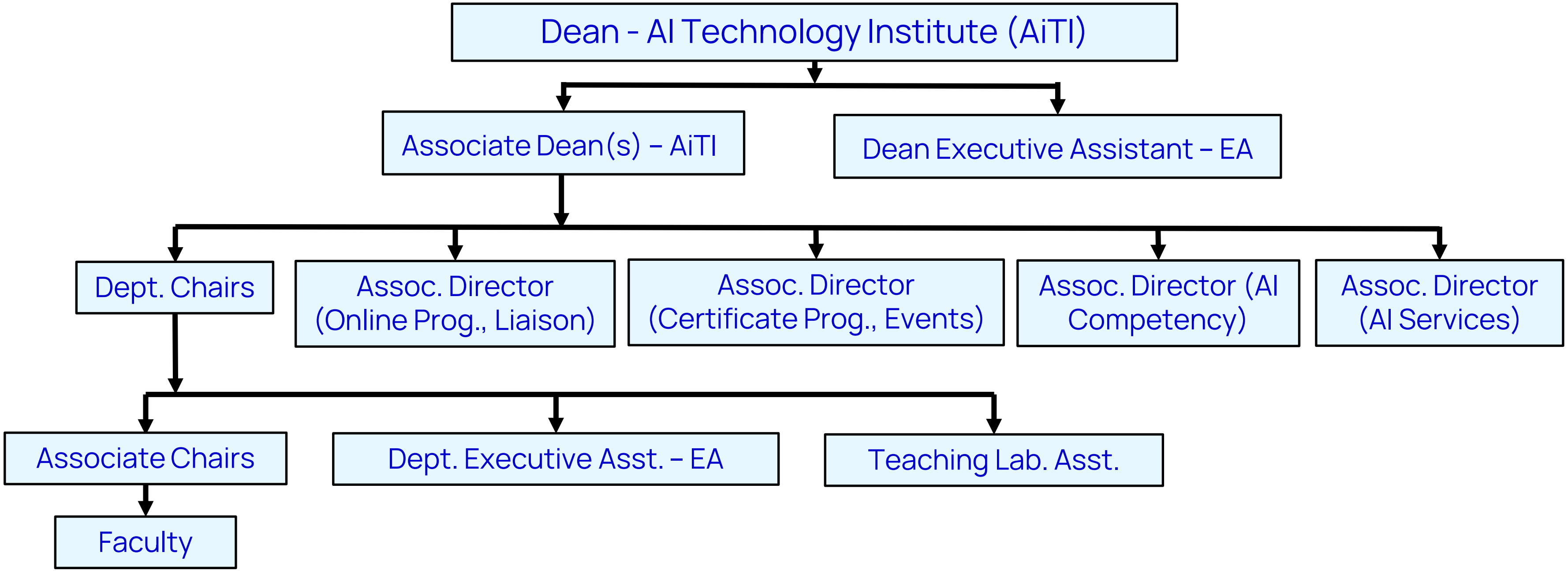
In-house AI computing facility

AiTI – Organizational Structure Overview



AiTI is organized into three functional pillars: **People** (Faculty, Recruitment Committee, Board of Studies, Students, External Users), **Administration** (Dean's Office, Departments/Clusters, Chairs & Executive Staff), and **Infrastructure** (High-End Nvidia Servers, Research Labs, Teaching Labs, Classrooms).

AiTI – Efficient Organizational Hierarchy



The Dean leads through Associate Deans who oversee Department Chairs, Associate Directors for Online Programs, Certificate Programs, AI Competency, and AI Services — cascading to Associate Chairs, Faculty, and Teaching Lab Assistants. The structure is designed to be **efficient and cost-optimal**.

AiTI – Clusters

AiTI's academic clusters are designed to be **futuristic**, capturing the full spectrum of science, technology, and application. Clusters will evolve into full Departments over time.

- **Information Engineering (INE)**
- **Applied AI (AAI)**
- **Language Technology & Cognitive Science (LCS)**
- **Cyber Defence Engineering (CDE)**
- **Robotics & Autonomous Systems (RAS)**
- **Cyber-Physical Systems (CPS)**

AI – Instant & Broad Publicity

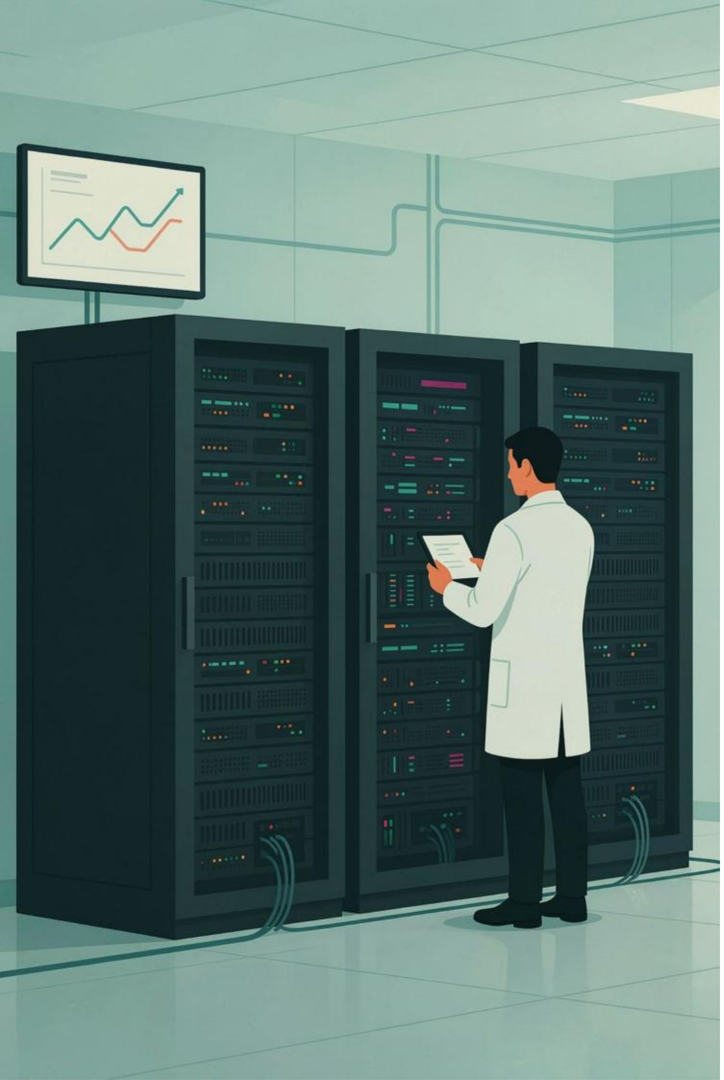
Prof. Mohanty's international stature has brought immediate visibility to AiTI. Featured on **Republic TV's R. Education Conclave** (10 May 2026) on the topic *"Role of AI in Healthcare"* with the headline: **"Future of Healthcare is Intelligent." Link**: https://www.youtube.com/watch?v=h31oCA5Xo1E

AiTI Recognized: Best Place to Study AI

Tech AI Magazine Award

AiTI has been awarded **"Best Place to Study AI"** by Tech AI Magazine — a testament to the rapid impact and recognition achieved under Prof. Mohanty's founding leadership.

This recognition reflects the institute's commitment to AI-native education, cutting-edge infrastructure, and industry-aligned programs from day one.



Founding Director: Smart Electronic Systems Lab (SESL)

UNIVERSITY OF NORTH TEXAS

SEP 2004 – PRESENT

Built a Nationally Recognized Center

Founded and grew SESL into a leading research lab in AI, IoT, Cybersecurity & Smart Systems.

Major Funding & Global Collaborations

Secured significant external research grants and built international research partnerships.

Advanced Infrastructure

Developed HPC and specialized lab infrastructure; mentored students into academic and industry leaders.

Professor & Academic Leader at UNT

COMPUTER SCIENCE & ENGINEERING

2004 – PRESENT

Strategic Leadership

Provided academic and administrative leadership in strategy, governance, curriculum, and student success.

ABET Accreditation

Led ABET accreditation and continuous quality improvement initiatives for Computer Engineering.

Faculty Recruitment

Played a key role in recruiting ~50 faculty through strategic search efforts over two decades.

Personnel Affairs Chair

Chaired Personnel Affairs Committee, overseeing tenure, promotion, and performance reviews.

IEEE Global Leadership

Board of Governors

Elected Member, IEEE Consumer Technology Society BoG (2019–2021)

Editor-in-Chief

IEEE Consumer Electronics Magazine — transformed from −$150K to +$150K surplus; IF from 0 to 4

TCVLSI Chair

Led IEEE-CS Technical Committee on VLSI with 1,000+ members and 100+ senior leaders

Founding iSES Chair

Founded IEEE iSES, establishing its governance, financial model, and strategic direction

Global Academic Leadership Roles

01

General Chair – Multiple IEEE International Conferences

Including ICCE 2018 (Las Vegas), ISVLSI 2019 (Miami), iSES 2021 (Jaipur), IFIP-IoT 2023 (DFW)

02

Founding Steering Committee Chair – IEEE iSES

Established governance, financial model, and long-term strategic direction since 2015

03

Editor-in-Chief – IEEE Consumer Electronics Magazine

2016–2021; Section EiC – Springer Nature Computer Science Journal (2020–present)

04

Leadership – IEEE ISVLSI, OCIT, IFIP Conferences

Steering Committee roles across multiple flagship IEEE and IFIP conferences

Budget Management Excellence

₹50Cr

AiTI Annual Budget

50 faculty, 10 staff, 3 programs, 6 depts, 10 labs & HPC infrastructure

$0.5M

IEEE Conference Impact

Annual economic impact overseeing 10 IEEE-CS-TCVLSI sponsored conferences

$160K

ICCE 2018 Surplus

Highest surplus in a decade from a single IEEE CE Society flagship conference

$300K

Magazine Turnaround

IEEE CE Magazine: from −$150K deficit to +$150K surplus over 5 years

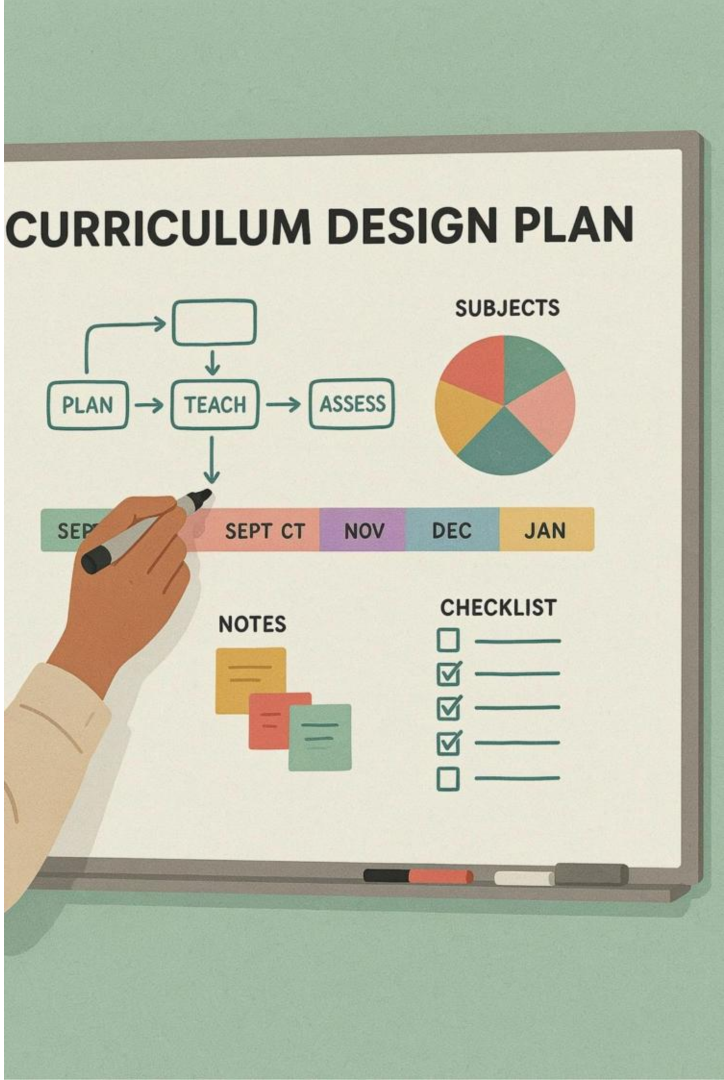
Curriculum Development

AI Programs – SRM-AP

Initiated B.Tech, M.Tech & Ph.D. AI programs at AiTI. Formed Board of Studies (BoS) and developed complete Outcome-Based Education (OBE) curricula and syllabi for all programs.

Computer Engineering – UNT

Founding faculty for CE at UNT in 2004. Established B.S., M.S., and Ph.D. programs. Developed full curricula and syllabi, achieving ABET accreditation from the very first term.



Outcome-Based Education (OBE)

Extensive experience developing OBE curricula across **USA, Middle East, and India**. OBE is a student-centric model where curriculum, teaching, and assessments are designed to achieve specific, measurable skills.

1

SMART Targets

Specific, Measurable, Achievable, Result-oriented, Time-bound learning goals

2

CO-PO Mapping

Course Outcomes (CO) and Program Outcomes (PO) mapped for measurable alignment

3

BLOOM Taxonomy

Structured lesson plans and assessments to encourage critical thinking at all levels

Learning Management Systems & Accreditation

LMS Experience

Experienced with LMS platforms for creating, delivering, and tracking educational materials:

- **USA:** Canvas, Blackboard
- **India:** Google Classroom
- **Recommended OBE-compatible:** Camu Digital Campus, VidyalayaLMS

Accreditation Experience

- **ABET:** Multiple decades, UNT & TIEC (USA)
- **AICTE:** Quality assurance for Indian engineering programs
- **NAAC:** Teaching-learning, infrastructure & research benchmarks
- **NBA:** Technical and professional program quality assessment

Personnel & Team Management

Founding Dean -- AiTI

Managing 2 Associate Deans, 6 HoCs, 50 faculty, 10 staff; ~240 students per year intake across B.Tech/M.Tech/Ph.D.

Founding Director -- Smart Electronics Lab

Managed 2 Co-Directors, 10 faculty, 2 staff; ~40 students per year intake across B.S/M.S/Ph.D.

Personnel Affairs Chair, Associate Chair, and Professor

Managed 40 faculty, 5 staff; ~500 students per year intake across B.Tech/M.Tech/Ph.D.

IEEE BoG Member

Persistent driver of budget, publications, and conferences at IEEE Consumer Electronics Society.

Conference Steering Chair

Managing organizing committees for IEEE ISVLSI, iSES, and OCIT annually since 2006.

Faculty Recruitment

AiTI – SRM-AP

Recruiting ~50 faculty per year as Founding Dean

UNT – 2 Decades

Helped recruit ~100 faculty as senior faculty member

Search Process

Prepares recruitment flyers, advertises in multiple media, directly reaches qualified candidates

Faculty Mentoring & Evaluation

Global Mentoring

Mentored faculty worldwide including Dr. Hui Zhao (UNT), Dr. Madhavi Ganapathiraju (Pitt), Deepak Puthal (Newcastle, UK), and Debanjan Das (IIIT Naya Raipur). Activities include co-authoring articles, organizing journal special issues, and joint conference organization.

Evaluation & Governance

As Dean AI, developed performance metrics covering Teaching, Research, and Service. As PAC Chair at UNT/CSE (multiple terms since 2010), presided over tenure, promotion, annual reappointment, and merit evaluations for the entire department.

Staff Management & Faculty Grievance

UNT

Instrumental in recruiting and mentoring several staff members over multiple decades.

IEEE

Recruited and mentored staff in capacity as EiC and IEEE Board of Governors member.

SRM-AP

Recruiting and mentoring staff as Founding Dean of AiTI.

Faculty Senate & Grievance

Served on UNT Faculty Senate and multiple faculty grievance committees as a senior faculty member.

Professional Leadership – Editorial Roles

Editor-in-Chief

IEEE Consumer Electronics Magazine (2016–2021)

Section Editor

Springer Nature Computer Science (2020–present)

IEEE Transactions on Big Data

Editorial Board (2018–present)

ACM JETC

Journal on Emerging Technologies in Computing Systems (2016–present)

Also serving on: IEEE Potentials Magazine, IEEE Trans. Nanotechnology, IET Circuits Devices & Systems, Elsevier VLSI Integration Journal, ASP JOLPE, and more.

Conference Chairmanship Highlights

2018 – Las Vegas

General Chair, 36th IEEE ICCE (co-located with CES) — generated $160K surplus

2019 – Miami

General Chair, 18th IEEE-CS ISVLSI Annual Symposium on VLSI

2021 – Jaipur

General Chair, 7th IEEE iSES International Symposium on Smart Electronic Systems

2022 – Bhubaneswar

General Chair, 20th OITS International Conference on Information Technology (OCIT)

2023 – DFW

General Chair, 6th IFIP International Internet of Things Conference

2025 – Fort Worth

Chair, Workshop on Quantum Solutions, 26th IEEE WoWMoM

Awards & Honors

PROSE Award 2016

Best Textbook in Physical Sciences & Mathematics — Mixed-Signal System Design

IEEE CE Society Outstanding Service Award

2020 – Leadership contributions to IEEE CT Society

Fulbright Specialist Award

U.S. Dept. of State Bureau of Educational & Cultural Affairs, 2021

IEEE Chester Sall Award

2020 – Best paper in IEEE Trans. Consumer Electronics

UNT Toulouse Scholars Award

2016–17 – Sustained excellence in scholarship and teaching

More Awards & Recognition

21
Best Paper Awards

Across IEEE international conferences and journals

2018
TCVLSI Distinguished Leadership

IEEE-CS-TCVLSI Distinguished Leadership Award for services to IEEE and VLSI community

2017
STC Award of Merit

Society for Technical Communication — outstanding contributions to IEEE CE Magazine

Current Research Focus

Smart Healthcare & Agriculture

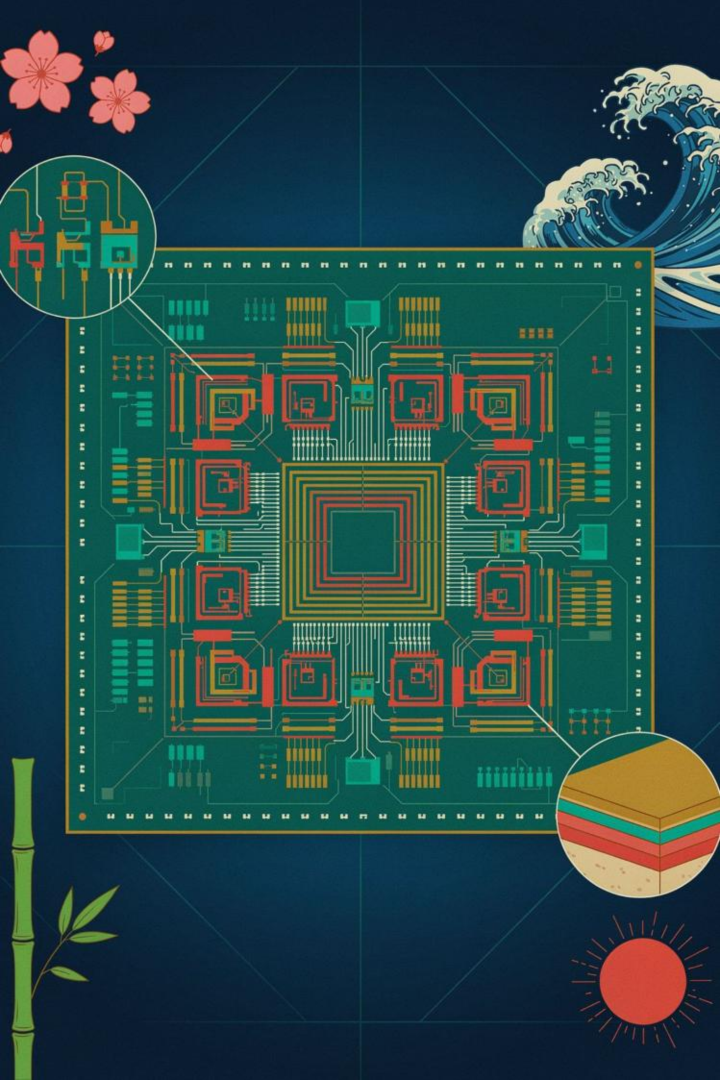
IoT-friendly approaches for smart healthcare monitoring and precision smart agriculture applications.

Smart Cities & Villages

IoT-enabled electronics powering intelligent urban and rural infrastructure systems.

Security-by-Design (SbD)

Embedding security into IoT and Cyber-Physical Systems (CPS) from the ground up.



Past Research Areas

Nanoscale Circuit Modeling

Modeling of nanoscale digital and analog/mixed-signal circuits for next-generation electronics.

Mixed-Signal Optimization

Methodologies for mixed-signal circuits and systems optimization — subject of award-winning textbook.

VLSI for Multimedia

VLSI architectures for multimedia signal processing, including pioneering digital watermarking chips (2004, 2006).

Research Achievements: By the Numbers

600
Research Articles
Mostly in IEEE transactions and conference proceedings

34
External Projects
NSF, SRC, US Air Force, NIDILRR, IUSTF, Mission Innovation

6
Patents Granted
Including first digital watermarking chip (2004) and secure digital camera (2004)

5
Books
Authored/co-authored, including PROSE Award-winning textbook

Google Scholar Impact

Prof. Dr. Saraju P. Mohanty

University of North Texas I Verified email at unt.edu

SMART HEALTHCARE

SMART AGRICULTURE

SECURITY BY DESIGN

20K

Total Citations

19,729 total; 12,580 since 2021

68

h-index

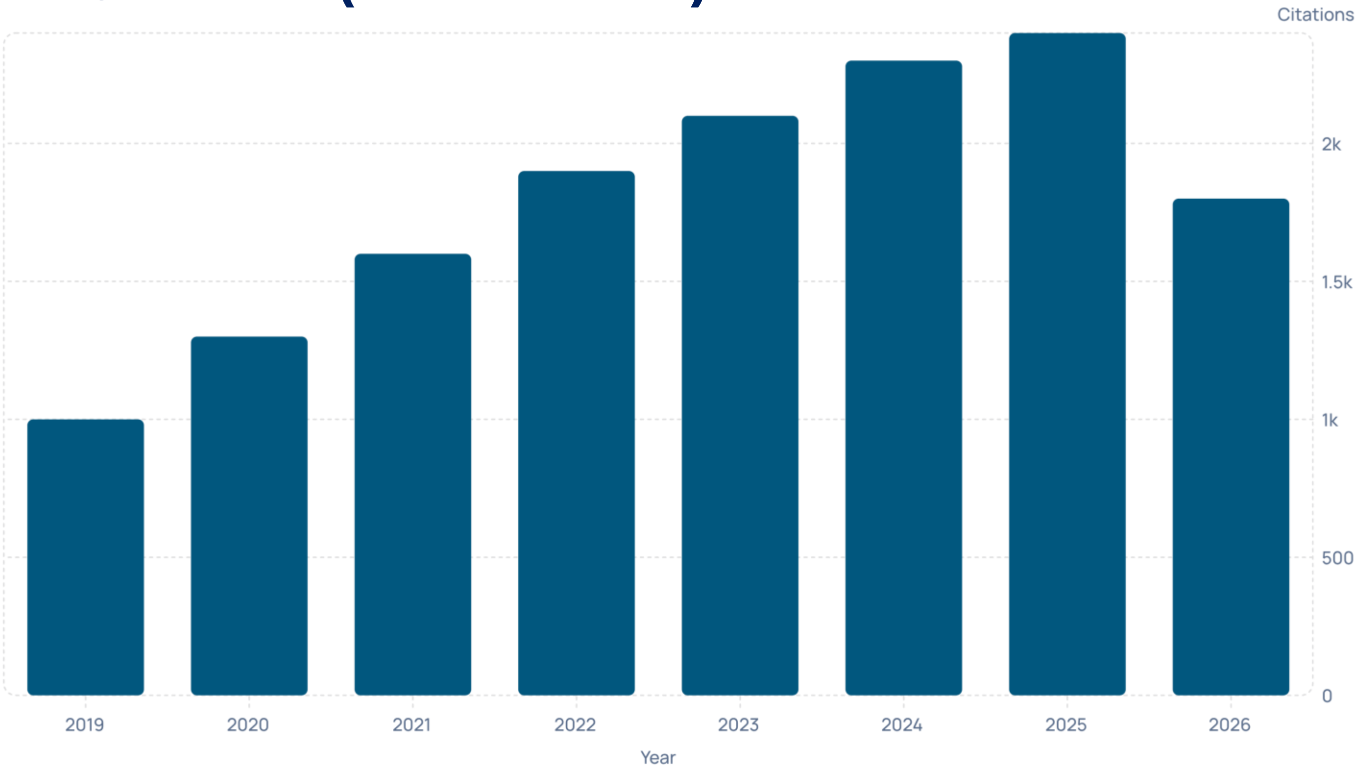
52 since 2021

341

i10-index

230 since 2021

Citation Growth (2019–2026)



Annual citations have grown steadily from ~1,000 in 2019 to a peak of ~2,400 in 2025, reflecting sustained and accelerating research impact. The 2026 figure represents a partial year.

Stanford Top 2% World Ranking

Top 2% Globally

As per Stanford University's ranking of the world's top scientists in **"Computer Hardware and Architecture"**:

2019	2020	2022
Ranked **30th** in the world	Ranked **25th** in the world	Ranked **14th** in the world

Top cited researcher on Google Scholar in: Smart Healthcare, Smart Agriculture, Smart Electronics, and Security by Design (SbD).

Significant International Visibility

34

Keynote Addresses

Delivered at IEEE International conferences worldwide

378

DBLP Co-Authors

Research collaborators around the globe

8000+

LinkedIn Connections

Global professional network

Active international connections through 3 decades of conference leadership, MOUs with global institutes for collaboration and student exchanges.

Teaching Achievements

Research Mentoring

3 post-docs, 21 Ph.D. dissertations, 29 M.S. theses, 41 undergraduate research projects

New Courses

Designed 4 new courses at UNT including Topics in Smart Electronics and Advanced Topics in VLSI Systems

ABET Accreditation

Founding CE faculty at UNT; achieved full ABET accreditation from first term; program ranked #73 by US News

Student Research Forum

Introduced SRF in IEEE ICCE, ISVLSI, and iSES to nurture future generation researchers

Ph.D. Mentoring – Selected Graduates

Ten advisees have received outstanding student/alumni awards at UNT. Selected Ph.D. graduates and their placements:

- **Recent Graduates (2019–2025)**

  S. Aarella (Austin College), L.S.T. Vangipuram (Texas A&M–Texarkana), A.K. Bapatla (Univ. of Central Missouri), L. Rachakonda (UNC Wilmington), I. Olokodana (IBM), M.A. Sayeed (Eastern New Mexico Univ.), V.P. Yanambaka (Central Michigan Univ.)

- **Earlier Graduates (2009–2018)**

  P. Sundaravadivel (UT Tyler), O. Okobiah (Samsung Semiconductor), G. Zheng (Analog Devices), O. Garitselov (Spectracom), D.V. Ghai (Pro-Chancellor, Oriental University, India)

Undergraduate & Graduate Student Mentoring

Undergraduate Mentoring

- 41 undergraduate research projects; TAMS high school students mentored annually
- Course Coordinator for hardware courses at UNT CSE for 18+ years
- CAD Tools Coordinator (Cadence) for College of Engineering — impacts 200+ students/year across 6 courses
- Developed undergraduate proposals for NSF, Dept. of Education, and federal agencies

Graduate Student Support

- Served on Graduate Studies Committee, CSE Department
- Instrumental in SACS accreditation of Computer Engineering Program
- Initiated student travel grants and best paper awards in IEEE conferences
- Consultant for TIEC — developed CS, CE, and IT curricula globally

Diversity & Inclusion in Mentoring

Half of all advisees are minorities in STEM. Prof. Mohanty has been a consistent champion of diversity across his career:

Women in STEM

Mentored 12+ women students; currently mentoring 4. Students attend Grace Hopper Celebration. Coordinated IEEE Women in Engineering activities at international conferences.

African-American Students

Advised 5 students including Dr. Oghenekarho Okobiah (15 articles, now at Intel) and Dr. Ibrahim Olokodana (10 articles, now at IBM).

Hispanic Students

Mentored 6 Hispanic students through graduate and undergraduate studies.

Professional Leadership – Full Portfolio

Editorial Leadership

EiC IEEE CE Magazine (2016–2021), Section EiC Springer Nature CS (2020–present), IEEE Trans. Big Data Steering Committee Secretary (2018–2000)

Steering Committee Roles

Founding Chair IEEE iSES (2015–present), Chair OCIT (2014–present), Vice-Chair ISVLSI (2016–present), Member DACS (2025–present)

Governance

IEEE CT Society Board of Governors (2019–2021), IEEE-CS TCVLSI Chair (2014–2018)

Editorial Board Memberships

Journal / Magazine	Period
IEEE Potentials Magazine	2025–present
IEEE Transactions on Big Data	2018–present
ACM Journal on Emerging Technologies in Computing Systems (JETC)	2016–present
IEEE Transactions on Nanotechnology	2017–present
IET Circuits, Devices and Systems	2014–present
Elsevier VLSI Integration Journal	2014–present
ASP Journal of Low Power Electronics	2011–present

Pioneer Research Milestones

1

2004

Designed the world's **first digital watermarking chip** and first **secure digital camera**

2

2006

Designed the world's **first low-power digital watermarking chip**

3

2015–Present

Leading IoT Security-by-Design, Smart Healthcare & Smart Agriculture research globally

4

2026

Building India's **first AI-Native University** as Founding Dean of AiTI, SRM-AP

A Legacy of Leadership, Research & Impact

🎓 Education

Ph.D. USF → 20+ years at UNT → Founding Dean AiTI, SRM-AP

🔬 Research

600 articles, 19K citations, h-index 68, Stanford Top 2%, 6 patents, 34 funded projects

🌐 Leadership

IEEE BoG, EiC, TCVLSI Chair, 34 keynotes, 378 global co-authors

🤝 Mentoring

21 Ph.D.s, 29 M.S., 41 undergrad projects; champion of diversity in STEM

Building India's first AI-Native University — where every graduate is AI-competent and every program is AI-integrated.